Auditing Tagging Process

A Framework for Validating Instructional Design and Event Data at the Center for Learning Analytics

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Abstract: This document presents a framework for auditing the instructional tagging process developed at the Center for Learning Analytics at the University of North Carolina at Chapel Hill. The process is organized into three phases. Phase 1 focuses on initial familiarization with the center's mission, research foundations, and instructional tagging framework. Phase 2 establishes the connection between course design and event data by examining course artifacts such as syllabi, schedules, and content clusters, and validating their alignment with tagging schema. Phase 3 involves the validation and analysis of tagged data through pivot tables and quantitative grids, enabling auditors to check consistency, temporal accuracy, and instructional intent alignment. Each phase includes detailed steps and screenshots from the tagging workbook.

Phase 1: Initial Familiarization with the Center for Learning Analytics at UNC

At this phase, the aim is to ensure that the person joining the labeling audit process understands where the research stands. The objective is to become familiar with both theory and practice, by understanding the files that will be worked with and how they are structured. This phase consists of three steps:

Step 1: Look at the official website of Center for Learning Analytics at the UNC School of Education. Reviewing the website will allow the person to learn about the center's mission, objectives, staff, and existing projects. This provides an understanding of the context in which the research is conducted.

Step 2: Read Plumley et al. (2024) – *Co-designing enduring learning analytics prediction and support tools in undergraduate biology courses.* This article is important because it presents a concrete example of how the center applies learning analytics

in higher education courses. By reviewing this study, the person will gain insight into the theoretical foundations, the co-design approach with instructors, and the practical results of using labeling and prediction models.

Step 3: Review slides 2 through 7 of the Instructional Tagging Introduction **to understand how the center organizes the tagging process. These slides present the logic of multimodal learning analysis, the role of course elements, and the key components of the tagging scheme. By studying these materials, you will gain a structured overview of how raw event data is enriched with course design elements.** Figure 1 shows t**he Instructional Design Grid. This grid organizes course materials by instructional purpose and timing (before, during, or after class). It makes visible how each resource is intended to support learning, providing a clear framework for tagging and analysis. Next, reviewing slides 9 and 10 will help to visualize what the final tagged product looks like, providing a clear picture of the expected outcomes of the tagging process.**

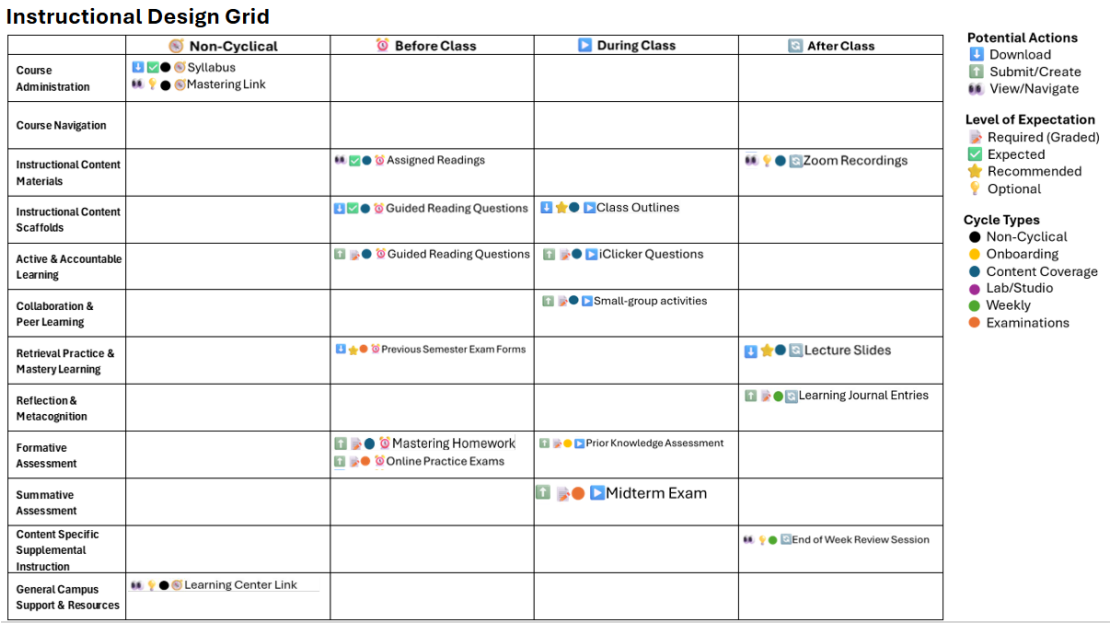


Figure ***1. Instructional Design Grid Initial***

Phase 2: Establishing the Link Between Course Design and Event Data

In this phase, work begins on the files for a particular course. The objective of this phase is to connect the elements of instructional design, such as the syllabus and activity schedule, with the labeling and event logging scheme. This is where the validation process begins, as the auditor starts to form their first impressions of how the labeling process was generated. In this phase, we will work on steps 4 to 7.

Step 4: At this stage, a course is selected to work on, for example PHYS115 from the fall semester of 2024. The idea is to read the syllabus, review the course schedule, and compare

them with the instructional design table (Figure 2). The objective of this step is to verify whether the planned instructional activities are aligned with the information in the official course documents. This is where the first comments and contributions are generated, with the auditor agreeing or disagreeing with the grid and helping to refine the outline so that the instructional intent is accurately captured.

PHYS115.001.FA24

	Non-Cyclical	Before Class	During Class	After Class
Course Administration	Schedule Syllabus Seating charts Agregate assigment	Plan of the Day		
Course Navigation	Book/E-book (OpenStax) Gradescope (Link to platform)			
Instructional Content Materials	Terms	Lesson Readings Additional Readings	Lecture Slides Lab Resources Video Studio Slides Complementary Materials	Class Recording
Instructional Content Scaffolds			Lecture Slides Tutorial Experiment Template	
Active & Accountable Learning			In-Class Activities (Poll Everywhere) Complementary Material Lab Instructions Lab Complementary Material	Homeworks (ExperTa) Lab Reports
Collaboration & Peer Learning			Studio Worksheets	Studio Submission Discussion
Retrieval Practice & Mastery Learning		Exam Resources	Studio Practice	Studio Worksheet Keys Practice Problems Post-Lecture Slides Lab Solutions
Reflection & Metacognition		Practice Exam Keys		Midterms Exams Form Midterms Exams Form Keys
Formative Assessment		Warm-up Assigment Practice Exams		
Summative Assessment			Midterms Exams Final Exam	
Content Specific Supplemental Instruction			Q&A End of Week Sessions	
General Campus Support & Resources				

Potential Actions
- Download
- Submit/Create
- View/Navigate

Level of Expectation
- Required (Graded)
- Expected
- Recommended
- Optional

Cycle Types
- Non-Cyclical
- Onboarding
- Content Coverage
- Lab/Studio
- Weekly
- Examinations

Figure 2. ***Instructional Design Grid for PHYS115 (Fall 2024), showing how course resources and activities are mapped across instructional categories and phases of the learning cycle.***

Step 5: Review slide 13 of the Instructional Tagging Introduction and then move on to the Excel Workbook with the data tagging already completed. The objective of this step is to ensure that all the elements found in the instructional design grid are reflected in the Excel sheet called "Common + Instructional Tagging", showing in Figure 3. The auditor must verify that the definitions and categories tagged match the grid. If there are any doubts or to verify understanding of the concepts used, you can read the process overview sheets (Figure 4) and instructional labeling diagram (Figure 5).

	A	B	C	D	E	F	G	H	I
3	Canvas	Post Class Lecture Slides	Download	Instructional Content Materials	Content Coverage	After Meeting	Recommended	No Stakes	0
4	Canvas	Article Reading	Download	Instructional Content Materials	Content Coverage	Before Meeting	Expected	No Stakes	0
5	OpenStax	Textbook Reading	View	Instructional Content Materials	Content Coverage	Before Meeting	Expected	No Stakes	0
6	GradeScope	Warm Up	Submit	Retrieval Practice & Mastery Learning	Content Coverage	Before Meeting	Required	Low-Stakes	5
7	ExpertTA	Homework	Submit	Retrieval Practice & Mastery Learning	Content Coverage	After Meeting	Required	Mid-Stakes	15
8	GradeScope	Lecture Tutorial	Submit	Collaboration & Peer Learning	Content Coverage	During Meeting	Required	Mid-Stakes	20
9		In-Class Workbook		Active & Accountable Learning	Content Coverage	During Meeting	Expected	No Stakes	0
10	Poll Everywhere	Poll Everywhere	Submit	Active & Accountable Learning	Content Coverage	During Meeting	Expected	No Stakes	0
11	Canvas	Schedule	Download	Course Administration	Non-Cyclical	Non-Cyclical	Expected	No Stakes	0
12	Canvas	Syllabus	Download	Course Administration	Non-Cyclical	Non-Cyclical	Expected	No Stakes	0
13	UNC Check In/ Canvas	Attendance	Post	Course Administration	Weekly	During Meeting	Expected	Low-Stakes	5
14	Canvas	Practice Exam Material	Download	Formative Assessment	Examinations	Before Meeting	Expected	No Stakes	0
15	Canvas	Midterm Exam	Submit	Summative Assessment	Examinations	During Meeting	Expected	High-Stakes	30
16	Canvas	Final Exam	Submit	Summative Assessment	Examinations	During Meeting	Required	High-Stakes	25
17		ignore							
18	Canvas	Cumilative Grade Totals	View	Course Administration	Non-Cyclical	Non-Cyclical	Recommended	No Stakes	0
19	Canvas	Course Announcements	View	Course Administration	Weekly	Non-Cyclical	Recommended	No Stakes	0
20	Canvas	Practice Exam Keys		Reflection & Metacognition	Examinations	Before Meeting	Optional	No Stakes	0
21	Canvas	Midterm Exam Keys		Reflection & Metacognition	Examinations	After Meeting	Optional	No Stakes	0
22	Canvas	Final Exam Keys		Reflection & Metacognition	Examinations	After Meeting	Optional	No Stakes	0
23	Canvas	Reading Links		Instructional Content Materials	Content Coverage	Before Meeting	Expected	No Stakes	0
24	Canvas	Lecture Recording Link		Instructional Content Materials	Content Coverage	Before Meeting	Expected	No Stakes	0
25	Canvas	Exam Bubble Sheets		Course Administration	Non-Cyclical	Non-Cyclical	Expected	No Stakes	0
26	Canvas	Course Navigation		Course Navigation	Non-Cyclical	Non-Cyclical	Optional	No Stakes	0
27	Canvas	Discussion		Collaboration & Peer Learning	Non-Cyclical	Non-Cyclical	Optional	No Stakes	0
28	Canvas	Astronomy Survey		Course Navigation	Non-Cyclical	Non-Cyclical	Optional	No Stakes	1
29									
30									
31									

Figure 3. ***Excerpt from the "Common + Instructional Tagging" sheet in the Instructional Tagging Workbook.***

	Tagging Process	Description
2	1) Instructional Intent Design Grid	Utilizing the syllabus, course schedule and objects list, populating course content materials to the instructional intent, intended timing, and cycle type. Content should be articulated utilizing color coding and mapping of content to their grid location
3	2) Common+Instructional Tagging	Similarly to the Instructional Intent Design Grid, utilizing the syllabus to populate course content material to into this tab. Within this tab more specifics can be gathered here incorporating further information such as "delivery platform", "stakes", "% of final grade", etc. This information will look to be granular to the course level and grouping content as necessary based on the Key Course Artifacts.
4	3) Cluster Topics	Mapping of content topics to Content Cluster labels. Content organizers and assessment mappings are documented.
5	4) Course Meetings List	Lists out all course meeting dates and times, labeling each with appropriate meeting number and the content cluster labels. More specific information such as attendance expectations & focal exam, The purpose is to provide a documentation of all course meetings to help determine whether an event is on-time, delayed, etc.
6	5) Object Names to CIT	Robert will provide the Object Type, Name and Action Type, as well as other pertinent information from the object files documents for each class. each Object will be mapped to the Content Cluster, Organization, and assessment Mapping
7	6) Interaction Date Lookup	Lists out all of the calendar dates during the course semester, indicating whether the date corresponds to a focal meeting
8	7) Tag Events	The application of look-ups from files from tabs 2, 3, and 5 to a long-form event file. This exists separately as it is a very large file on its own.
9	7a) Tag Events Readme	Descriptions of variable headers in 7) Tagged Events
10	8) Instructional Intent Quant Grid	Presents quantitative data of common name events, organized by instructional intent categories. This shows the number of unique students access vs no access, total number of access, average number of access, and percentages. Allows for instructors to investigate the breakdown of access for students during the course of the semester.
11	9) Quant Sandbox	Pivot table linked to 7) Tagged Events for exploratory analysis

Figure 4. ***Process Overview sheet from the Instructional Tagging Workbook, summarizing the main steps and definitions used in the tagging process.***

Tag Type	Tag Values	Tag Description
Action	Download	Acquire a copy of files (e.g., .doc, .ppt, .pdf) from the digital platform for use on local machine
Action	Post	Create a discussion or message
Action	Reply	Create a response to a discussion or message
Action	Submit	Submit a completed submission to an asisgnment
Action	Update	Edit an assignment submission or discussion contribution
Action	Attach	Include a file on a generated submission, discussion contribution, etc.
Action	View	Load or navigate to web pages(e.g., course pages, discussion pages, external links)
Expectation	Required	Actions directly related to a graded activity.
Expectation	Expected	Mandatory but not graded (e.g., pre-class readings, downloading files needed for graded activites).
Expectation	Recommended	Strongly encouraged but not required.
Expectation	Optional	Extra/enrichment material for deeper learning or review.
Cycle Phase	Non-Cyclical	Resources available throughout the course, not tied to phases.
Cycle Phase	Before Meeting	Resouces/activities intended to be engaged with before the related class meeting
Cycle Phase	During Meeting	Resouces/activities intended to be engaged with during the related class meeting
Cycle Phase	After Meeting	Resouces/activities intended to be engaged with after the related class meeting has occurred
Cycle Type	Non-Cyclical	Resources and activities that are not related to a specific class meeting or topic.
Cycle Type	Onboarding	Introductory activities to familiarize students with course expectations.
Cycle Type	Content Coverage	Instructional sessions focused on delivering and exploring course material.
Cycle Type	Lab/Studio	Hands-on, practical sessions for applying concepts in real-world scenarios.
Cycle Type	Weekly	Recurring sessions/assignments that maintain course rhythm and reinforce learning.
Cycle Type	Examinations	High-stakes, summative assessments.
Instructional Intent	Course Administration	Documents and tools that provide structural organization and course policies.
Instructional Intent	Course Navigation	Viewing of course site pages and external links.
Instructional Intent	Instructional Content Materials	Core instructional materials that provide foundational knowledge on course topics.
Instructional Intent	Instructional Content Scaffolds	Resources designed to guide students in understanding and organizing new information.
Instructional Intent	Active & Accountable Learning	Activities intended to encourage engagement where students apply and explore concepts interactively.
Instructional Intent	Collaboration & Peer Learning	Opportunities for students to engage in peer-supported learning experiences.
Instructional Intent	Retrieval Practice & Mastery Learning	Strategies that enhance retention and recall of key concepts through structured practice.
Instructional Intent	Reflection & Metacognition	Resources and activities that encourage students to reflect on their learning process.
Instructional Intent	Formative Assessment	Low-stakes assessments providing students with feedback on their understanding.
Instructional Intent	Summative Assessment	High-stakes evaluations used to measure students' mastery of the course material.
Instructional Intent	Content Specific Supplemental Instruction	Supplemental instruction of course specific content provided by the learning center, teaching-assistants, peer-mentors, etc.
Instructional Intent	General Campus Support & Resources	University-based services to develop students' academic writing, and research.
Stakes	No Stakes	Not direcly related to a graded activity. This applies to viewing or downloading instructional materials unless the event is direcity related to participation points or grades are explicitly assign
Stakes	Low-Stakes	Very low % of course grade, often evaluated based on participation not correctness
Stakes	Mid-Stakes	Meaningful, but not substancial % of course grade. Evaluated based on correctness.
Stakes	High-Stakes	Substancial portion of final course grade. Evaluated for correctness.

Figure 5. ***Instructional Tagging Schema sheet, detailing the categories and labels***

applied to course elements during the tagging process.

Step 6: The auditor should now familiarize themselves with the "Content Cluster" sheet in the labeling workbook (see Figure 6) and compare it with the course syllabus. By reviewing this sheet, the auditor ensures that the labeling structure accurately reflects the duration and sequence of the course. This step also allows for the identification of discrepancies between the intended organization of the course and the way events are grouped in the labeling process, providing an early opportunity to refine the cycle.

	A	B	C	D
1	content_topic	content_cluster	content_organizers	content_assessments
2	Introduction to Organic Chemistry	L01	Unit 01	Exam 1, Exam 4
3	Bonding and Hybridization	L02	Unit 01	EOW01, Exam 1, Exam 4
4	Nomenclature and Physical Propertie	L03	Unit 02	EOW01, Exam 1, Exam 4
5	Conformational Isomers and Strain E	L04	Unit 02	EOW02, Exam 1, Exam 4
6	Substituted Cyclohexanes and Chairs	L05	Unit 02	EOW02, Exam 1, Exam 4
7	Chirality Part 1	L06	Unit 02	EOW03, Exam 1, Exam 4
8	Chirality Part 2 / Review	L07	Unit 02	EOW03, Exam 1, Exam 4
9	Bronsted-Lowry Acids and Bases, Res	L08	Unit 03	Exam 1, Exam 4
10	Acid-Base Trends, Lewis Acids and Ba	L09	Unit 03	EOW04, Exam 2, Exam 4
11	Introduction to Alkenes and Thermod	L10	Unit 04	EOW05, Exam 2, Exam 4
12	Kinetics and Intro to Electrophilic Add	L11	Unit 04	EOW05, Exam 2, Exam 4
13	Electrophilic Addition Reactions 2	L12	Unit 04	EOW06, Exam 2, Exam 4
14	Electrophilic Addition Reactions 3	L13	Unit 04	EOW06, Exam 2, Exam 4
15	Electrophilic Addition Reactions 4	L14	Unit 04	EOW07, Exam 2, Exam 4
16	Electrophilic Addition 5 - Alkyne Reac	L15, L16	Unit 04	EOW08, Exam 3, Exam 4
17	Nucleophilic Substitution Reactions 1	L17	Unit 05	EOW08, Exam 3, Exam 4
18	Nucleophilic Substitution Reactions 2	L18	Unit 05	Exam 3, Exam 4
19	Solvent Effects and Elimination React	L19	Unit 05	EOW09, Exam 3, Exam 4
20	Elimination Reactions 2	L20	Unit 05	EOW09, Exam 3, Exam 4
21	Substitution vs. Elimination	L21	Unit 05	EOW10, Exam 3, Exam 4
22	Substitution with Alcohol Derivatives	L22	Unit 05	EOW10, Exam 3, Exam 4
23	Alcohols, Ethers, and Epoxides	L23	Unit 06	EOW11, Exam 3, Exam 4
24	Organometallic Reagents	L24	Unit 07	EOW11, Exam 3, Exam 4
25	Radical Reactions 1	L25	Unit 07	Exam 4
26	Radical Reactions 2 and Final Review	L25	Unit 07	Exam 4

Figure 6. ***Content Clusters sheet from the Instructional Tagging Workbook, showing how individual content topics are grouped into clusters and aligned with course organizers and assessments.***

Step 7: Now we move on to reviewing the tagged events sheet of the document in Excel to understand how student interactions with course materials were tagged. This sheet (Figure 7) documents raw event data, such as downloads, submissions, or views, and connects them to the categories created in the design grid. It is important to note that this process was performed manually by researchers, meaning that human judgment influenced the classification of events. By examining this sheet, the auditor gains insight into the practical challenges of tagging, such as consistency, accuracy, and the need for clear guidelines in order to tag the number of rows presented. At this stage, it would be useful to have an explanatory video illustrating how this labeling process was carried out, to clarify how the person reviewed the names of the listed objects and tagged them.

context_sis_source_id	context_section_id	object_name	object_type	object_action_type	content_common_name	content_instructional_intent	content_cycle_type	content_cycle_phase	content_expectation
CHEM261.001.SP24	47276	24SP261_Syllabus_Eskew.docx	file	file-download	Syllabus	Course Administration	Non-Cyclical	Non-Cyclical	Expected
CHEM261.001.SP24	47270	Pre-Course Assessment	wiki_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47270	course-home	course_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47270	Welcome to CHEM261	discussion	discussion-accessed	Announcements	Course Administration	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47270	Gradescope 1.3	externalTool	externalTool-accessed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47270	Your Instructor	wiki_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47276	If You're Absent	wiki_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47276	Canvas Resources for Students	wiki_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47270	Unit 1 Supplemental Instructional Videos	wiki_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47276	course-home	course_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47276	Homepage	wiki_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47270	Course Reserves 1.0	externalTool	externalTool-accessed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47276	Getting Help	wiki_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47276	Welcome to CHEM261	discussion	discussion-accessed	Announcements	Course Administration	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47276	Pre-Course Assessment - Version 1	quiz	quiz-accessed	Pre Course Assessment	Formative Assessment	Onboarding	Non-Cyclical	Required
CHEM261.001.SP24		Pre-Course Assessment - Version 1	quiz	quiz-submitted	Pre Course Assessment	Formative Assessment	Onboarding	Non-Cyclical	Required
CHEM261.001.SP24	47270	Getting Help	wiki_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47270	Unit 1 Learning Objectives	wiki_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47276	Gradescope 1.3	externalTool	externalTool-accessed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47270	Writing Center	externalTool	externalTool-accessed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47276	course-grades	course_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47276	Syllabus	wiki_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47270	Syllabus	wiki_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47270	24SP261_Syllabus_Eskew.docx	file	file-download	Syllabus	Course Administration	Non-Cyclical	Non-Cyclical	Expected
CHEM261.001.SP24	47270	Pre-Course Assessment - Version 1	assignment	assignment-accessed	Pre Course Assessment	Formative Assessment	Onboarding	Non-Cyclical	Required
CHEM261.001.SP24	47270	Welcome to CHEM261	discussion	discussion-accessed	Announcements	Course Administration	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24		Pre-Course Assessment - Version 1	assignment	assignment-submission-	Pre Course Assessment	Formative Assessment	Onboarding	Non-Cyclical	Required
CHEM261.001.SP24		Pre-Course Assessment - Version 2	assignment	assignment-submission-	In-Class Activity	Active & Accountable Learning	Content Coverage	During Meeting	Required
CHEM261.001.SP24	47270	University Policies	wiki_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47270	course-topics	course_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47276	Gradescope 1.3	externalTool	externalTool-accessed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47276	Canvas Resources for Students	wiki_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47276	Your Instructor	wiki_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47270	Learning Center	externalTool	externalTool-accessed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47270	Learning Center	externalTool	externalTool-accessed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47270	Learning Center	externalTool	externalTool-accessed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47270	Pre-Course Assessment - Version 2	assignment	assignment-accessed	In-Class Activity	Active & Accountable Learning	Content Coverage	During Meeting	Required
CHEM261.001.SP24	47270	Course Reserves 1.0	externalTool	externalTool-accessed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended
CHEM261.001.SP24	47270	Pre-Course Assessment - Version 1	quiz	quiz-accessed	Pre Course Assessment	Formative Assessment	Onboarding	Non-Cyclical	Required
CHEM261.001.SP24	47276	24SP261_Syllabus_Eskew.docx	file	file-download	Syllabus	Course Administration	Non-Cyclical	Non-Cyclical	Expected
CHEM261.001.SP24	47270	Homepage	wiki_page	page-viewed	Course Navigation	Course Navigation	Non-Cyclical	Non-Cyclical	Recommended

3) Content Clusters | 4) Course Meetings List | 5) Object Names to CIT | 6) Interaction Date Lookup | 7) Tagged Events | Detail1 | 8) Inst. Int

Figure 7. ***Tagged Events sheet from the Instructional Tagging Workbook, showing how raw student interactions with course materials are manually categorized and aligned with instructional tags.***

Phase 3: Validation and Analysis of Tagged Data

At this stage, the focus is on reviewing and confirming the accuracy of the tagging scheme through detailed data analysis. The auditor works with pivot tables and tagged data sets to explore how student interactions with elements are represented, identifying inconsistencies and verifying that the tagging reflects the course design.

Step 8: Examine the pivot table found in the "Instructional Intent Quant Grid" in the workbook. This table (Figure 8) shows the count of events by object, providing a percentage of how students interacted with different elements of the course. The auditor should carefully review the grid to identify patterns, inconsistencies, or anomalies that stand out, such as unusually high or low engagement with specific materials.

Sum of event_count	Column Labels								
Row Labels	Non-cyclical	pre_fdoc	before_multi_mtg	before_one_mtg	before_days	during_dayofclass	after_one_mtg	after_multi_mtg	post_ldoc
Non-Cyclical									
Non-Cyclical									
Course Navigation									
Course Navigation									
accessed	66.19%	2.93%	8.14%	3.73%	1.41%	1.89%	3.11%	9.44%	3.16%
Course Administration									
Syllabus									
download	73.40%	20.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	6.60%
Announcements	81.15%	9.71%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	9.13%
Schedule	90.77%	8.88%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.36%
Seat Assignments									
download	99.92%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.08%
Exam Regrade Request									
download	98.86%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.14%
Assignment Template									
download	99.84%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.16%
General Campus Support & Resources									
Study Strategies									
download	100.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Onboarding									
Non-Cyclical									
Formative Assessment									
Pre Course Assessment									
accessed	0.00%	0.00%	0.00%	0.00%	7.53%	4.24%	32.22%	55.80%	0.22%
submission_created									
Pre-Course Assessment - Version 1	0.00%	0.00%	0.00%	0.00%	4.41%	4.87%	72.62%	18.10%	0.00%
Pre-Course Assessment - Version 2	0.00%	0.00%	0.00%	0.00%	0.24%	0.00%	0.00%	99.76%	0.00%
Content Coverage									
Non-Cyclical									
Reflection & Metacognition									
Additional Problems Key									
download	98.41%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.59%
Retrieval Practice & Mastery Learning									
Additional Problems									
download	99.65%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.35%
Before Meeting									
Instructional Content Scaffolds									

4) Course Meetings List | 5) Object Names to CIT | 6) Interaction Date Lookup | 7) Tagged Events | Detail1 | 8) Inst. Int. Quant Grid | 7a) Ta

Figure 8. ***Instructional Intent Quantitative Grid from the tagging workbook***

Step 9: Check temporal tag accuracy by working with actual dates and examples of how to create the pivot table in 8) Inst. Quant Grid to understand the results on that pivot table… which elements go to in rows, columns and values or filters.

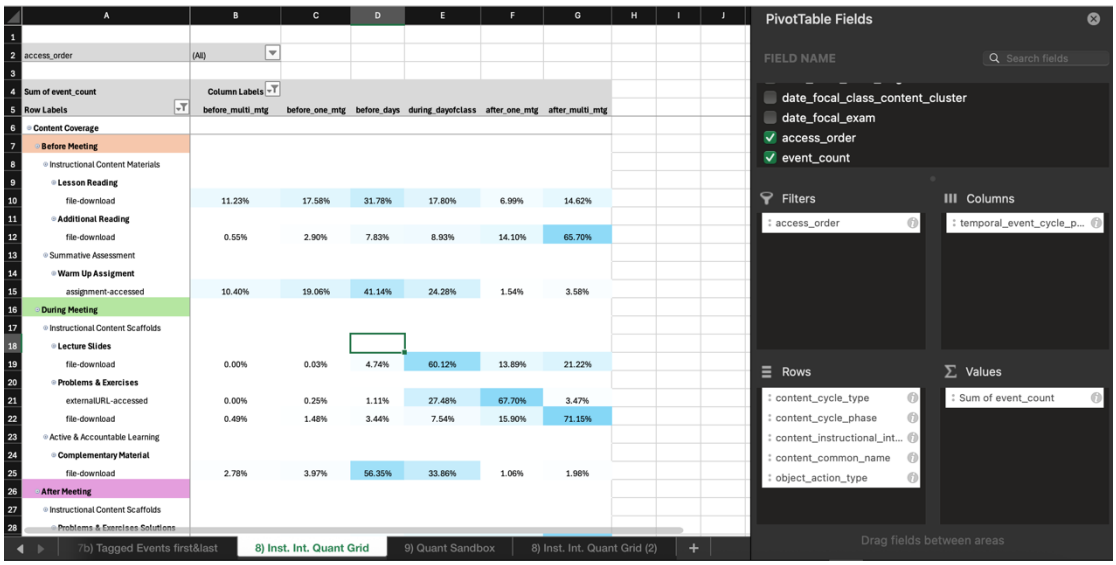
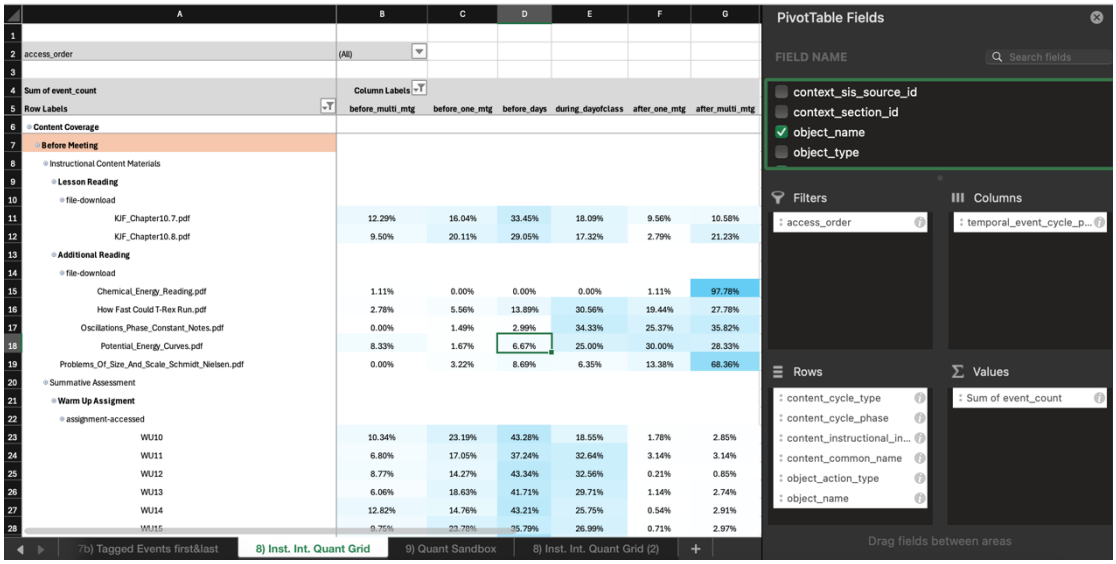


Figure 9. ***Multi-Level Material Access Patterns by Category and Type***



access_order	(All)					
Sum of event_count	Column Labels					
Row Labels	before_multi_mtg	before_one_mtg	before_days	during_dayofclass	after_one_mtg	after_multi_mtg
Content Coverage						
Before Meeting						
Instructional Content Materials						
Lesson Reading						
file-download						
KJF_Chapter10.7.pdf	12.29%	16.04%	33.45%	18.09%	9.56%	10.58%
KJF_Chapter10.8.pdf	9.50%	20.11%	29.05%	17.32%	2.79%	21.23%
Additional Reading						
file-download						
Chemical_Energy_Reading.pdf	1.11%	0.00%	0.00%	0.00%	1.11%	97.78%
How Fast Could T-Rex Run.pdf	2.78%	5.56%	13.89%	30.56%	19.44%	27.78%
Oscillations_Phase_Constant_Notes.pdf	0.00%	1.49%	2.99%	34.33%	25.37%	35.82%
Potential_Energy_Curves.pdf	8.33%	1.67%	6.67%	25.00%	30.00%	28.33%
Problems_Of_Size_And_Scale_Schmidt_Nielsen.pdf	0.00%	3.22%	8.69%	6.35%	13.38%	68.36%
Summative Assessment						
Warm Up Assigment						
assignment-accessed						
WU10	10.34%	23.19%	43.28%	18.55%	1.78%	2.85%
WU11	6.80%	17.05%	37.24%	32.64%	3.14%	3.14%
WU12	8.77%	14.27%	43.34%	32.56%	0.21%	0.85%
WU13	6.06%	18.63%	41.71%	29.71%	1.14%	2.74%
WU14	12.82%	14.76%	43.21%	25.75%	0.54%	2.91%
WU15	9.75%	23.78%	35.79%	26.99%	0.71%	2.97%

Figure 10. ***Additional Object Name Granularity Added to Existing Categories***

These two pivot tables demonstrate Step 9's approach to understanding temporal tag accuracy through progressive data granularity. Figure **9** shows materials already organized by multiple hierarchical categories (content_cycle_type, content_cycle_phase, content_instructional_intent, content_common_name, etc.) displaying their access patterns across time phases. Figure **10** builds on this same categorization structure by adding object_name as an additional row field, revealing individual files like "KJF_Chapter10.7.pdf" and specific assignments (WU10-WU15) within the existing category framework. This demonstrates how adding one more level of detail helps identify which specific files within each category have problematic temporal patterns requiring investigation.

Step 10: Check temporal tag accuracy by creating a new sheet with first and last access data that cause questions/concerns.

access_order	first						
temporal_event_cycle_phase	(All)						
Sum of event_count	**Column Labels**						
Row Labels	**Exam1**	**Exam2**	**Exam3**	**FinalExam**	**L01**	**L01, L02**	**L02**
Problems & Exercises							
externalURL-accessed							
L01							
Conceptual Survey (complete in first 30 min of Studio 1)	0.00%	0.00%	0.00%	0.00%	45.00%	0.00%	55.00%
L26							
Conceptual Survey (complete in the first 30 min of Studio 26)	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
file-download							
L02							
ScalingPonderables.pdf	9.09%	0.00%	0.00%	0.00%	1.52%	12.12%	19.70%
L03							
MALTHUSIAN GROWTH.pdf	6.25%	0.00%	0.00%	0.00%	0.00%	6.25%	0.00%
L14							
Torque_Ponderables.pdf	0.00%	11.54%	19.23%	0.00%	0.00%	0.00%	0.00%
L17							
Cheat_sheet_on_Work_energy_relation.docx	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%

Figure 1111. ***First Access Timing Distribution for Problems & Exercises Materials***

This pivot table shows when different Problems & Exercises files are first accessed across content clusters and exam periods.

Step 11: Check instructional intent accuracy by analyzing temporal mismatches that combine action type (access, download, submit) with temporal context, focusing on actions occurring after the expected phase (especially after multiple meetings) to identify when timing and behavior type together change the instructional function.

object_typ	object_action_typ	content_common_name	content_instructional_intent	content_cycle_typ	content_cycle_pha	temporal_event_cycle_phas
externalURL	externalURL-accessed	Lab Instructions	Active & Accountable Learning	Lab/Studio	During Meeting	before_days
file	file-download	Lab Instructions	Active & Accountable Learning	Lab/Studio	During Meeting	before_days
externalURL	externalURL-accessed	Lab Instructions	Active & Accountable Learning	Lab/Studio	During Meeting	during_dayofclass
file	file-download	Lab Instructions	Active & Accountable Learning	Lab/Studio	During Meeting	during_dayofclass
externalURL	externalURL-accessed	Lab Instructions	Active & Accountable Learning	Lab/Studio	During Meeting	after_one_mtg
file	file-download	Lab Instructions	Active & Accountable Learning	Lab/Studio	During Meeting	after_one_mtg
file	file-download	Lab Instructions	Active & Accountable Learning	Lab/Studio	During Meeting	after_multi_mtg
externalURL	externalURL-accessed	Lab Instructions	Active & Accountable Learning	Lab/Studio	During Meeting	after_multi_mtg
file	file-download	Lab Instructions	Active & Accountable Learning	Lab/Studio	During Meeting	before_multi_mtg

Figure 1212. ***Lab Instructions with Static Intent Labels Across Multiple Access Times***

As this screenshot shows, Lab Instructions materials all labeled “Active & Accountable Learning” despite being accessed at different times - before class, during class, and after class (highlighted in yellow). The expected phase is “During Meeting” but actual access spans multiple temporal phases. This demonstrates how static instructional intent labels fail to capture that materials serve different functions when accessed outside their intended timeframe.

Step 12: Intended phase vs % of temporal phase for first access

To begin our audit process, we compared the intended instructional phases (before, during, after meeting, non-cyclical) (see figure 1) with the temporal phases observed in student data. Using pivot tables, we calculated the proportion of first and total accesses for each phase. For example, while the syllabi indicated that 36.8% of student interaction should occur after meeting, the data shows 48.41% of actual interactions fell into this phase. (see figure 2)

Intended phase		
Row Labels	**Count of context_sis_source_id**	
After Meeting	15586	37%
Before Meeting	7018	17%
During Meeting	9952	23%
Non-Cyclical	9794	23%
Grand Total	**42350**	

Figure 1 Students' total count based on intended phase

Temporal event phase		
Row Labels	**Count of context_sis_source_id**	
after_multi_mtg	16487	0.4841086
after_one_mtg	4015	
before_days	2479	
before_multi_mtg	2257	
before_one_mtg	1827	
during_dayofclass	2979	
non_cyclical	10181	
post_ldoc	1496	
pre_fdoc	629	
Grand Total	**42350**	

Figure 2 Students' total count based on temporal event cycle phase

This discrepancy does not by itself explain why the difference occurred, but it **signals a potential mismatch** between design and behavior. Rather than concluding that students postponed engagement, we use this phase-level comparison as an initial screen to flag areas that merit closer review.

To formalize this check, we compute a mismatch ratio (temporal ÷ intended). Ratios substantially greater than 1 (e.g., after meeting≈ 1.1) indicate over-representation in each phase, while ratios substantially lower than 1 (e.g., during meeting≈ 0.32) suggest under-representation relative to expectations. These ratios highlight which phases and resources should be examined further.

Finally, we determined that the temporal cycle phase list should include all individual events, not just aggregated categories. This more granular view allows us to trace mismatches back to specific resources (e.g., additional readings, warm-ups, assignments) and better understand where student behaviors diverge from the intended design.

Step 13 What signs in the data might indicate inaccurate labels for specific objects?

To move from aggregate patterns to specific resources, we examined Additional Reading in PHYS114. According to the instructional tagging schema, this content was intended to support preparation before meeting. However, the temporal phase data (see figure 3) shows that most accesses occurred after meeting (268 out of 450 total events, compared to only 85 before meeting).

This misalignment suggests that the current labeling may not accurately reflect how students are engaging with this resource. Instead of serving as pre-class preparation, students appear to be using Additional Reading primarily for post-class review. Such evidence indicates a potential labeling mistake and raises the question of whether the intentional intent should be revised.

Screenshots of these distributions can help illustrate the mismatch. By grounding the review in object-level data, we can better decide whether this resource's classification should remain unchanged or be updated to reflect actual student usage.

content_common_name	Additional Reading
Temporal event phase	
Row Labels	**Count of context_sis_source_id**
after_multi_mtg	268
after_one_mtg	54
before_days	51
before_multi_mtg	12
before_one_mtg	22
during_dayofclass	40
post_ldoc	3
Grand Total	**450**

Figure 3 Temporal event phase of additional reading

Step 15: How should we account for different population sizes vs it simply being an infrequently utilized object?

See this link.

Phase 4: Automated Validation Using R

This phase introduces R-based automated validation procedures that scale the manual audit process established in Phase 3. The R scripts systematically calculate validation metrics, generate distribution visualizations, and produce standardized check tables that can be applied consistently across multiple courses.

Objectives of Phase 4

1) Automate the calculation of student population benchmarks and engagement proportions
2) Generate standardized validation checks for both cyclical and non-cyclical objects
3) Produce replicable visualizations that reveal engagement patterns and anomalies
4) Establish data-driven thresholds for classification decisions

Step 16: Setting Up the R Environment and Loading Data

Begin by loading the required R packages and importing the tagged events data. The script uses `tidyverse` for data manipulation and `here` for reproducible file paths.

r

```
library(tidyverse)

library(here)


raw_data <- read_csv(here("Week 11", "PHYS114_001_SP24_TEST_summary_report.csv"))
```

Key Requirements:

1) The input data must be the "Tagged Events" sheet exported as CSV
2) Data should include all necessary fields: `access_order`, `content_cycle_phase`, `object_name`, `date_focal_class_content_cluster`, `temporal_event_cycle_phase`, etc.
3) Filter for `access_order %in% c("first", "first&last")` to focus on initial engagement patterns

Step 17: Establishing Student Population Benchmark

The first critical step is identifying the total student population, which serves as the denominator for all engagement calculations.

Non-Cyclical Objects as Population Estimator:

We use non-cyclical objects (materials not tied to specific modules) to estimate the student population. The logic is that the maximum number of unique students who accessed any non-cyclical object represents the course enrollment.

r

```r
# Aggregate first access counts for non-cyclical objects
data_noncyclical_total <- raw_data |>
  filter(access_order %in% c("first", "first&last"),
                 content_cycle_phase == "Non-Cyclical") |>
  group_by(object_name, object_type) |>
  summarise(event_count = sum(event_count), .groups = "drop")

# Identify maximum as population benchmark
max_unique_users <- max(data_noncyclical_total$event_count)
```

Result for PHYS114: `max_unique_users = 426 students`

Key Output Files:

1) `data_noncyclical_totaltest.csv`: Total first access per non-cyclical object
2) `data_cyclical_totaltest.csv`: Total first access per cyclical object
3) `noncyclical_event_distribution.png`: Histogram visualization
4) `cyclical_event_distribution.png`: Histogram visualization

Step 18: Focal Cluster Concentration Analysis

This step determines whether objects show concentrated access patterns in specific course modules or dispersed access throughout the semester.

For Non-Cyclical Objects:

Calculate the proportion of each object's total access that occurred during each focal content cluster:

r

```
data_noncyclical_focal_prop <- data_noncyclical_focal |>
  group_by(object_name, object_type) |>
  mutate(total_event_count = sum(event_count),
         focal_proportion = event_count / total_event_count) |>
  ungroup()
```

Identify the focal cluster with maximum proportion for each object:

r

```
data_noncyclical_max_focal <- data_noncyclical_focal_prop |>
  group_by(object_name, object_type) |>
  slice_max(focal_proportion, n = 1, with_ties = FALSE)
```

For Cyclical Objects:

Identify which focal cluster had the highest first access for each cyclical object:

r

```
data_cyclical_focal <- raw_data |>
  filter(access_order %in% c("first", "first&last"),
         content_cycle_phase != "Non-Cyclical") |>
  group_by(object_name, date_focal_class_content_cluster) |>
  summarise(event_count = sum(event_count), .groups = "drop") |>
  group_by(object_name) |>
  mutate(focal_proportion = event_count / sum(event_count)) |>
  slice_max(focal_proportion, n = 1)
```

Key Metric: Mean Focal Proportion

Calculate the mean of maximum focal proportions for cyclical objects:

r

```
threshold_X <- mean(data_cyclical_focal$focal_proportion)
# Result: 0.6215426 (62.15%)
```

This threshold represents typical access concentration for module-specific materials and becomes the benchmark for validating non-cyclical objects.

Key Output Files:

1) `data_noncyclical_focal_proportionstest.csv`: All focal proportions for non-cyclical objects
2) `data_noncyclical_max_focaltest.csv`: Maximum focal cluster per non-cyclical object
3) `data_cyclical_focal_maxtest.csv`: Maximum focal cluster per cyclical object
4) `noncyclical_focal_proportion_distribution.png`: Histogram
5) `cyclical_max_focal_proportion_distribution.png`: Histogram

Step 19: Temporal Phase Validation

This step validates whether objects are labeled with the correct instructional phase (Before/During/After Meeting) by comparing expected phases with actual student access timing.

Temporal Phase Aggregation:

Simplify the granular temporal phases into three broad categories:

r

```
data_cyclical_temporal <- raw_data |>
  mutate(
    temporal_phase_agg = case_when(
      temporal_event_cycle_phase %in% c("pre_fdoc", "before_multi_mtg",
                                        "before_one_mtg", "before_days") ~ "before",
      temporal_event_cycle_phase == "during_dayofclass" ~ "day_of",
```

```
        temporal_event_cycle_phase %in% c("after_one_mtg", "after_multi_mtg",
                                          "post_ldoc") ~ "after"
    )
  )
```

Object-Level Temporal Check:

For each object, identify which temporal phase had the highest proportion of first access, then compare with the labeled `content_cycle_phase`:

r

```
data_cyclical_object_temporal <- data_cyclical_temporal |>
  group_by(object_name, temporal_phase_agg) |>
  summarise(event_count = sum(event_count), .groups = "drop") |>
  group_by(object_name) |>
  mutate(temporal_proportion = event_count / sum(event_count)) |>
  slice_max(temporal_proportion, n = 1) |>
  mutate(
    expected_phase_agg = case_when(
        content_cycle_phase == "Before Meeting" ~ "before",
        content_cycle_phase == "During Meeting" ~ "day_of",
        content_cycle_phase == "After Meeting" ~ "after"
    ),
    Object_level_check = temporal_phase_agg == expected_phase_agg
  )
```

Common Name-Level Temporal Check:

Aggregate across all objects of the same type to identify systematic labeling issues:

r

```
data_cyclical_common_temporal <- data_cyclical_temporal |>
  group_by(content_common_name, temporal_phase_agg) |>
  summarise(event_count = sum(event_count), .groups = "drop") |>
  group_by(content_common_name) |>
  mutate(temporal_proportion = event_count / sum(event_count)) |>
```

```
  slice_max(temporal_proportion, n = 1) |>
  mutate(Common_name_check = temporal_phase_agg == expected_phase_agg)
```

Findings from PHYS114:

Materials with systematic temporal misalignment (`Common_name_check = FALSE`):

1) **Additional Reading** (labeled "During Meeting", actual "after") - 5 objects
2) **Studio Worksheet** (labeled "Before Meeting", actual "after") - 22 objects
3) **Studio Data Sheet** (labeled "During Meeting", actual "after") - 8 objects
4) **Studio Instructions** (labeled "During Meeting", actual "after") - 9 objects
5) **Practice Exam** (labeled "After Meeting", actual "before") - 11 objects

These patterns suggest either:

1) **Labeling errors** - The intended phase was misidentified
2) **Behavioral patterns** - Students consistently use materials differently than intended
3) **Hybrid functions** - Materials serve different purposes at different access times

Key Output Files:

1) `data_cyclical_object_temporaltest.csv` - Object-level temporal validation
2) `data_cyclical_common_temporaltest.csv` - Common name-level temporal validation
3) `data_cyclical_combined_checkstest.csv` - Combined validation results

Step 20: Non-Cyclical Object Validation

Apply the threshold derived from cyclical objects (62.15%) to validate non-cyclical object classifications.

Two-Condition Validation Logic:

r

```
data_noncyclical_checked <- data_noncyclical_total |>
  left_join(data_noncyclical_max_focal, by = c("object_name", "object_type")) |>
  mutate(
    Object_level_check = case_when(
      # Condition 1: If content_cluster label exists, verify match
      !is.na(content_content_cluster) ~
```

```
            max_focal_cluster == content_content_cluster,

        # Condition 2: If no content_cluster label, verify dispersion
        is.na(content_content_cluster) ~
            max_focal_proportion < threshold_X
    )
  )
```

Interpretation of Results:

1) **PASS (TRUE)**: Object correctly classified as non-cyclical
    - Either labeled cluster matches maximum focal cluster
    - Or no cluster label AND max proportion < 62.15%
2) **FAIL (FALSE)**: Object requires review
    - Labeled cluster doesn't match actual maximum focal cluster
    - Or no cluster label BUT max proportion ≥ 62.15% (concentrated access suggests it should be cyclical)

Key Output File:

1) `data_noncyclical_checkedtest.csv` - Comprehensive validation results for all non-cyclical objects

Step 21: Interpreting Automated Validation Results

Categories of Validation Outcomes:

1) **Pass Both Checks (Object + Common Name)**
    - Material is correctly tagged
    - No action needed
2) **Fail Common Name Check**
    - Systematic issue affecting entire category
    - **Action**: Consider relabeling the common name itself
    - **Example**: Studio Worksheets labeled "Before Meeting" but accessed "after" across 22 objects
3) **Pass Common Name, Fail Object Check**
    - Individual tagging error or exception
    - **Action**: Investigate specific object for:
        - Schedule changes
        - Data quality issues
        - Genuine exceptions to typical pattern
4) **Non-Cyclical Concentration Issue**

    - Non-cyclical object with focal proportion ≥ 62.15%
    - **Action**: Consider assigning content_cluster label and reclassifying as cyclical

Decision Framework:

When validation checks fail, the auditor must determine:

1) Is this a **data quality issue**? (low enrollment, missing data)
2) Is this a **labeling error**? (wrong phase/cluster assigned)
3) Is this **unexpected student behavior**? (materials used differently than intended)
4) Is this a **legitimate exception**? (schedule change, optional material)

Key Deliverables from Phase 4

1) **CSV Output Files** (12 files per course):
    - Non-cyclical: total, focal, proportions, max focal, checked
    - Cyclical: total, focal max, object temporal, common temporal, combined
2) **Visualization Files** (4 PNG images):
    - Non-cyclical and cyclical event distributions
    - Non-cyclical and cyclical focal proportion distributions
3) **Summary Statistics**:
    - Student population benchmark
    - Threshold X (mean focal proportion for cyclical objects)
    - Counts of objects failing each validation check
4) **Flagged Objects Lists**:
    - Common names with systematic temporal misalignment
    - Individual objects with concentration/cluster mismatches
    - Non-cyclical objects potentially needing reclassification